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Course: DSA0602/ Data Handling and Visualization

List of Experiments — Sets 24 to 30 (Questions, R Code & Output)

Set 24: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Note: Question 2 requires a Category grouping and Question 4 requires a Price value, neither of which is present in the Quantity-only table above. Reasonable placeholder Category and Price columns (matching the values used earlier in this experiment set) are added below so both charts can be produced.

Question 1 (R)

In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

R Code:

```
df24 <- data.frame(
  ProductID = 1:5,
  ProductName = c("Product A","Product B","Product C","Product D","Product E"),
  Qty = c(250, 175, 300, 200, 220),
  Price = c(20, 15, 18, 25, 22),
  Category = c("Cat1","Cat2","Cat1","Cat2","Cat1")
)

ggplot(df24, aes(x = ProductName, y = Qty, fill = ProductName)) +
  geom_bar(stat = "identity", show.legend = FALSE) + theme_minimal() +
  labs(title = "Inventory Quantity by Product", x = "Product", y = "Quantity Available")
```

Output — Question 1:

Interpretation:

Product C holds the largest stock (300 units); Product B holds the least (175 units).

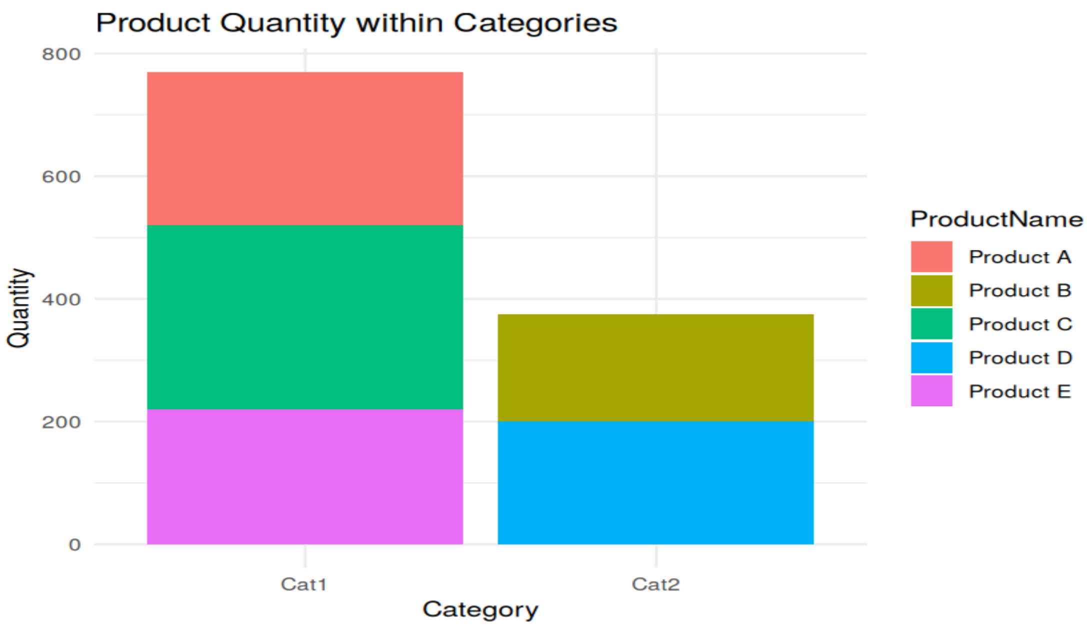
Question 2 (R)

In R, generate a stacked bar chart to show the quantity of each product within different product categories.

R Code:

```
ggplot(df24, aes(x = Category, y = Qty, fill = ProductName)) +
  geom_bar(stat = "identity") + theme_minimal() +
  labs(title = "Product Quantity within Categories", x = "Category", y = "Quantity")
```

Output — Question 2:



Interpretation:

Cat1 (Products A, C, E) holds more combined stock than Cat2 (Products B, D).

Question 3 (Tableau)

In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.

Steps to perform in Tableau:

- Connect Tableau to Product_Inventory.csv.
- Sheet 1: Product Name on Columns, SUM(Qty) on Rows — standard bar chart.
- Sheet 2: Category on Columns, SUM(Qty) on Rows, Product Name on Color — stacked bar chart.
- Combine both sheets on a Dashboard and add a Category filter applied to both.

Question 4 (R)

In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

R Code:

```
ggplot(df24, aes(x = Price, y = Qty)) +
  geom_point(size = 4, color = "darkred") + theme_minimal() +
  labs(title = "Product Price vs Quantity Available", x = "Price ($)", y = "Quantity")

cor(df24$Price, df24$Qty)
```

Output — Question 4:

Interpretation:

Insight: the correlation between Price and Quantity here is r ≈ -0.05 — essentially no linear relationship in this small sample, meaning stock levels are not simply driven by price. Product C (mid-priced) carries the most stock, while Product B (cheapest) carries the least, so restocking looks to be based on demand or category rather than price.

Set 25: Website Traffic Analysis

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

Question 1 (Tableau)

In Tableau, create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

Steps to perform in Tableau:

- Connect Tableau to Website_Traffic.csv.
- Drag Date to Columns (continuous, exact date) and Page Views to Rows.
- Change the mark type to Line, add a title 'Daily Page Views Trend', and label both axes.

Question 2 (R)

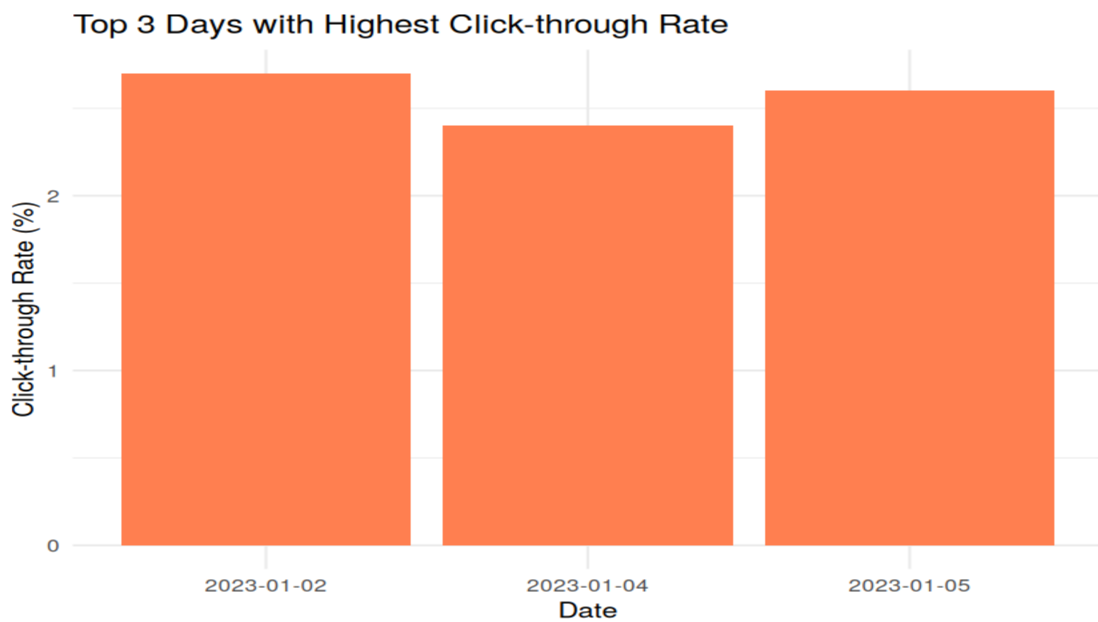
In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

R Code:

```
df25 <- data.frame(
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")),
  Views = c(1500, 1600, 1400, 1650, 1800), CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)
```

```
top_ctr <- df25 %>% arrange(desc(CTR)) %>% head(3)
ggplot(top_ctr, aes(x = as.factor(Date), y = CTR)) +
  geom_bar(stat = "identity", fill = "coral") + theme_minimal() +
  labs(title = "Top 3 Days with Highest Click-through Rate", x = "Date", y = "Click-through
Rate (%)")
```

Output — Question 2:



Interpretation:

Jan 2 (2.7%), Jan 5 (2.6%) and Jan 4 (2.4%) are the top 3 CTR days.

Question 3 (R)

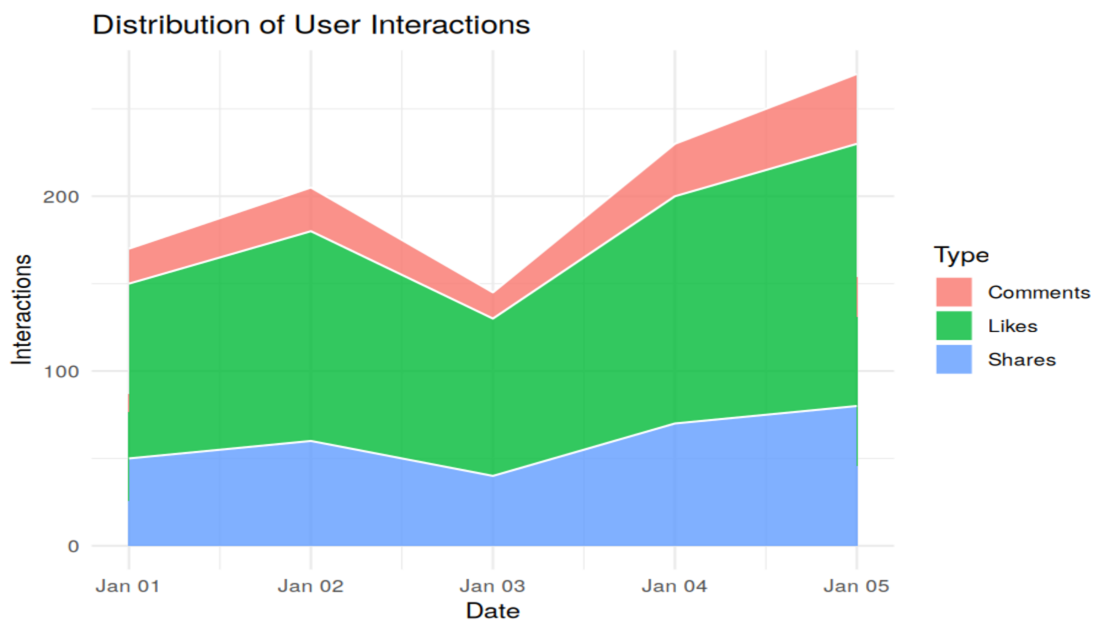
In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

R Code:

```
df25_int <- data.frame(
  Date = df25$Date, Likes = c(100, 120, 90, 130, 150),
  Shares = c(50, 60, 40, 70, 80), Comments = c(20, 25, 15, 30, 40)
)
df25_int_long <- pivot_longer(df25_int, cols = -Date, names_to = "Type", values_to = "Count")

ggplot(df25_int_long, aes(x = Date, y = Count, fill = Type)) +
  geom_area(alpha = 0.8, color = "white") + theme_minimal() +
  labs(title = "Distribution of User Interactions", x = "Date", y = "Interactions")
```

Output — Question 3:



Interpretation:

Likes consistently dominate the interaction mix, followed by shares and then comments, across all five days.

Question 4 (Tableau)

In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.

Steps to perform in Tableau:

- This requires a Country/City (traffic-source location) field, which is not present in the Website Traffic dataset — add a 'Source Country' column to the CSV to enable geocoding.
- Sheet 1 (Map): drag the geographic field to the view; Tableau auto-generates Latitude/Longitude. Put SUM(Page Views) on Size and/or Color.
- Sheet 2 (Bar chart): Source on Columns, SUM(Page Views) on Rows.
- Combine both on a Dashboard with a shared Source filter.

Set 26: Student Mini Data (Student_Mini_Data.csv)

Dataset: Student_Mini_Data.csv

Student_ID	Gender	Study_Hours	Attendance	Math_Score	Science_Score	Exam_Date
S01	Male	2.0	78	62	65	2025-01-10
S02	Female	3.5	90	80	85	2025-01-10
S03	Male	1.5	70	55	58	2025-02-12
S04	Female	4.0	95	90	92	2025-02-12
S05	Male	2.8	85	72	74	2025-03-15
S06	Female	3.0	92	82	86	2025-03-15

Question 1 (R)

Import the dataset in R. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.

R Code:

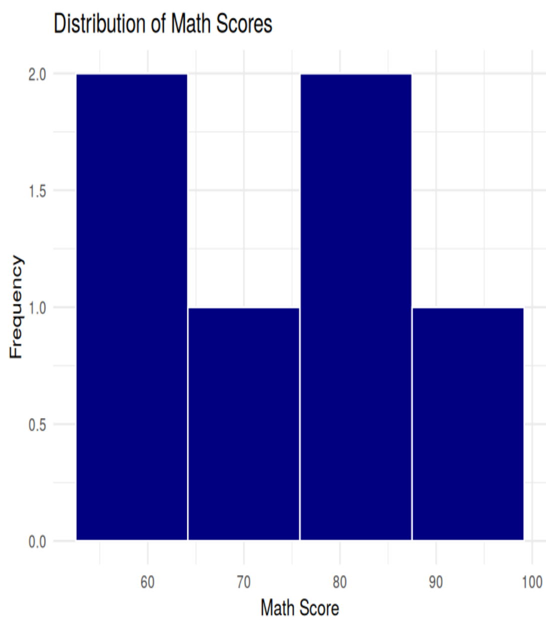
```
df26 <- data.frame(
  Student_ID = c("S01","S02","S03","S04","S05","S06"),
```

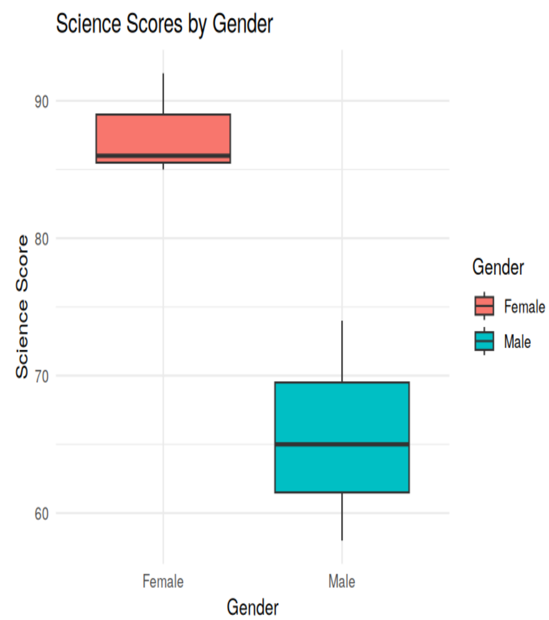
```
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Study_Hours = c(2.0, 3.5, 1.5, 4.0, 2.8, 3.0),
  Attendance = c(78, 90, 70, 95, 85, 92),
  Math_Score = c(62, 80, 55, 90, 72, 82),
  Science_Score = c(65, 85, 58, 92, 74, 86),
  Exam_Date = as.Date(c("2025-01-10","2025-01-10","2025-02-12","2025-02-12","2025-03-15","2025-
03-15"))
)

ggplot(df26, aes(x = Math_Score)) +
  geom_histogram(bins = 4, fill = "navy", color = "white") + theme_minimal() +
  labs(title = "Distribution of Math Scores", x = "Math Score", y = "Frequency")

ggplot(df26, aes(x = Gender, y = Science_Score, fill = Gender)) +
  geom_boxplot() + theme_minimal() +
  labs(title = "Science Scores by Gender", x = "Gender", y = "Science Score")
```

Output — Question 1:





Interpretation:

Interpretation: Math scores are spread fairly evenly from 55-90 with no single dominant bin. In the boxplot, Female students show a visibly higher median Science score (~85) than Male students (~65) with less spread, suggesting more consistent performance among the female students in this small sample.

Question 2 (R)

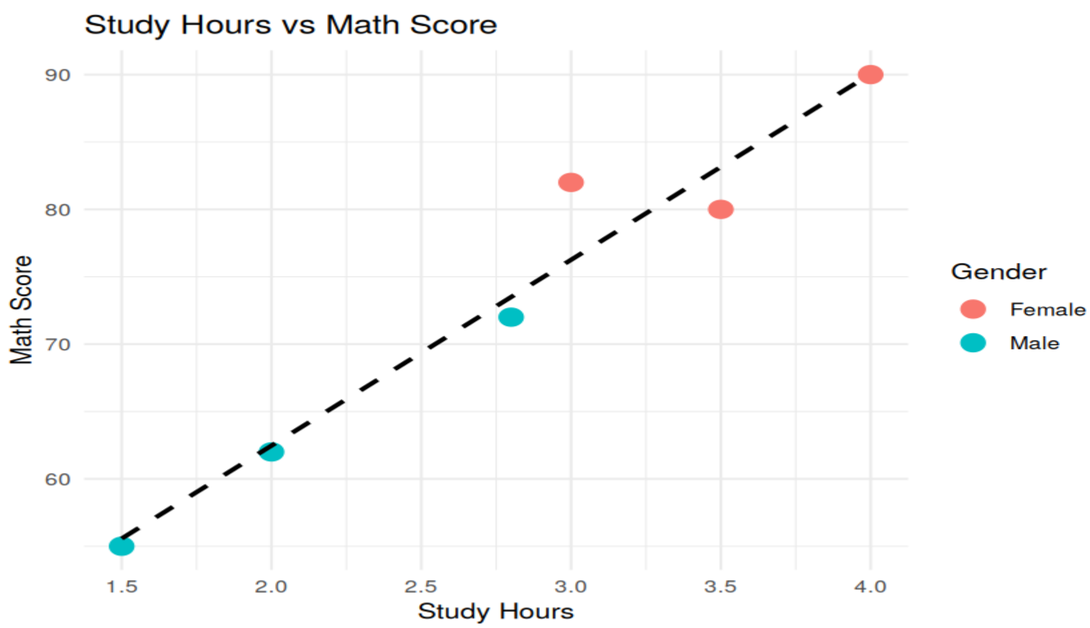
Create a scatter plot using R programming between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.

R Code:

```
ggplot(df26, aes(x = Study_Hours, y = Math_Score, color = Gender)) +
  geom_point(size = 4) +
  geom_smooth(method = "lm", se = FALSE, color = "black", linetype = "dashed") +
  theme_minimal() + labs(title = "Study Hours vs Math Score", x = "Study Hours", y = "Math
Score")

cor(df26$Study_Hours, df26$Math_Score)
```

Output — Question 2:



Interpretation:

The correlation between Study Hours and Math Score is r ≈ 0.97 — a strong positive linear relationship. Students who study more hours consistently score higher in Math, and this pattern holds across both genders.

Question 3 (R)

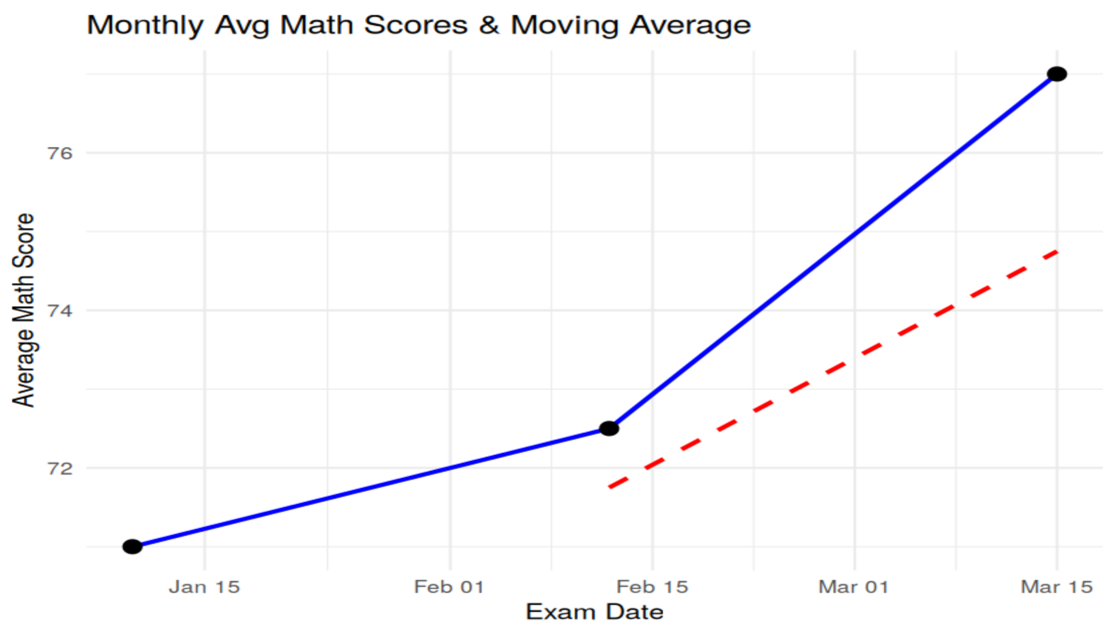
Using R, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.

R Code:

```
df26_agg <- df26 %>%
  group_by(Exam_Date) %>%
  summarize(AvgMath = mean(Math_Score)) %>%
  mutate(MA = rollmean(AvgMath, k = 2, fill = NA, align = "right"))

ggplot(df26_agg, aes(x = Exam_Date)) +
  geom_line(aes(y = AvgMath), color = "blue", linewidth = 1) +
  geom_point(aes(y = AvgMath), size = 3) +
  geom_line(aes(y = MA), color = "red", linetype = "dashed", linewidth = 1) +
  theme_minimal() + labs(title = "Monthly Avg Math Scores & Moving Average",
                         x = "Exam Date", y = "Average Math Score")
```

Output — Question 3:



Interpretation:

Average Math scores rise steadily across the three exam dates: 71.0 -> 72.5 -> 77.0. The 2-point moving average (red dashed line) confirms a consistent upward trend, indicating overall class improvement over time.

Question 4 (Tableau)

Import Student_Mini_Data.csv into Tableau. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for Exam_Date. Design a simple interactive dashboard.

Steps to perform in Tableau:

- Connect Tableau to Student_Mini_Data.csv.
- Sheet 1 (Bar): Gender on Columns, AVG(Math_Score) on Rows.
- Sheet 2 (Line): Exam_Date on Columns (continuous), AVG(Math_Score) on Rows.
- Drag Exam_Date to the Filters shelf (range or relative-date filter).
- Combine both sheets on a Dashboard and apply the Exam_Date filter to both.

Set 27: Patient Health Risk Analysis

Dataset: Patient Health

Patient_ID	Age	BMI	BP	Cholesterol
P1	25	22	120	180
P2	40	28	135	210
P3	55	30	145	240
P4	35	26	130	200
P5	60	32	150	260

Question 1 (R)

Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

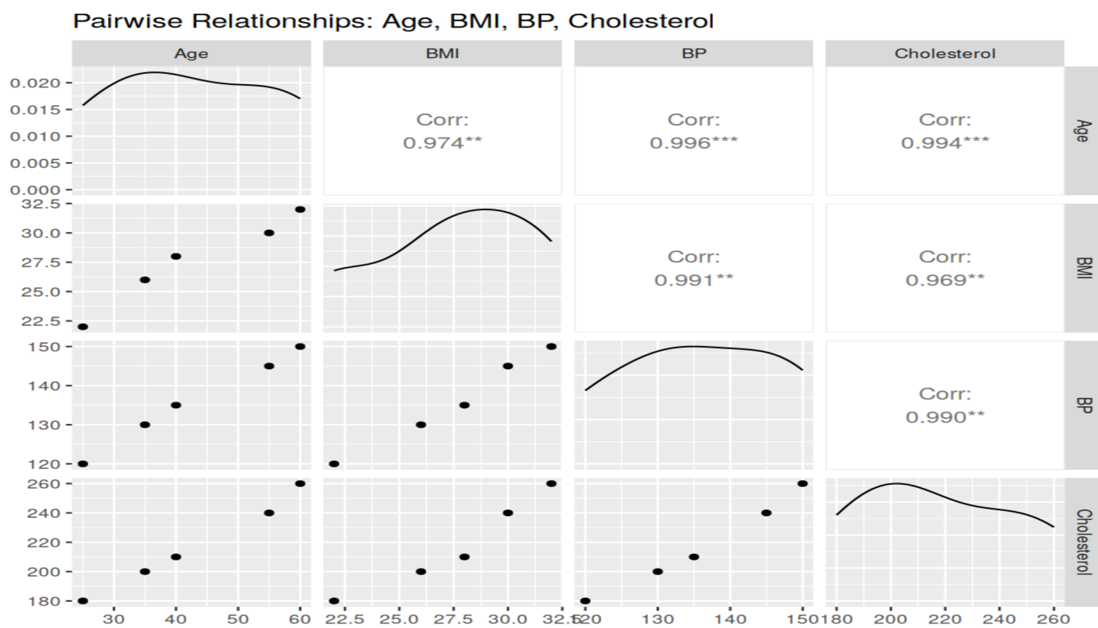
R Code:

```
library(GGally)
```

```
df27 <- data.frame(
  Patient_ID = c("P1","P2","P3","P4","P5"), Age = c(25,40,55,35,60),
  BMI = c(22,28,30,26,32), BP = c(120,135,145,130,150), Cholesterol = c(180,210,240,200,260)
)

ggpairs(df27[,2:5], title = "Pairwise Relationships: Age, BMI, BP, Cholesterol")
```

Output — Question 1:



Interpretation:

All four indicators are extremely strongly correlated with each other ($r > 0.96$ in every pair): as Age rises, BMI, BP, and Cholesterol all rise together in near-lockstep. There is no separate clustering — the five patients form a single, consistent risk gradient from P1 (lowest, youngest) to P5 (highest, oldest).

Question 2 (R)

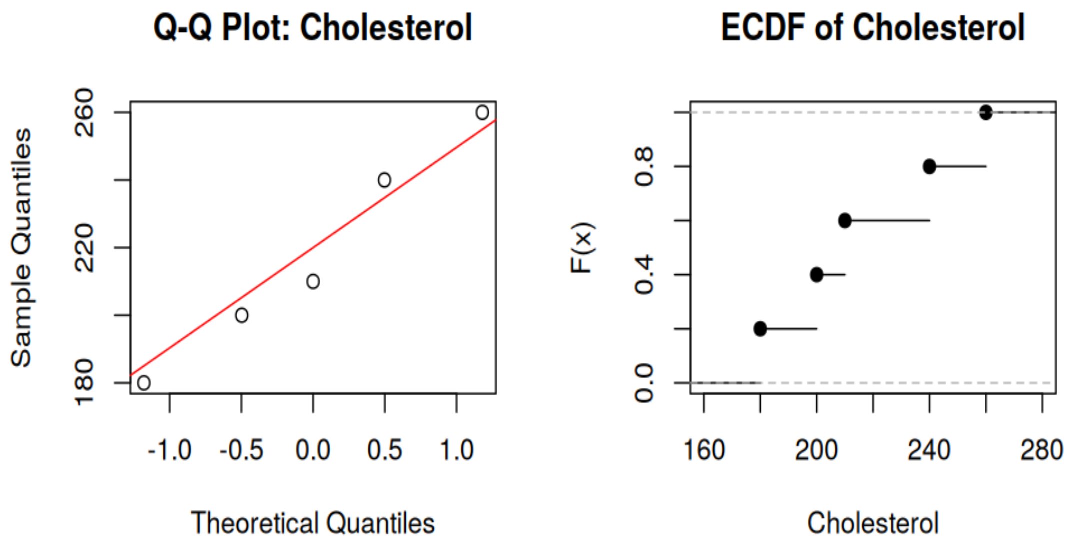
Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

R Code:

```
par(mfrow = c(1,2))
qqnorm(df27$Cholesterol, main = "Q-Q Plot: Cholesterol")
qqline(df27$Cholesterol, col = "red")
plot(ecdf(df27$Cholesterol), main = "ECDF of Cholesterol", xlab = "Cholesterol", ylab = "F(x)")
par(mfrow = c(1,1))

shapiro.test(df27$Cholesterol)
```

Output — Question 2:



Interpretation:

The Q-Q points fall close to the reference line, and a Shapiro-Wilk test gives W = 0.967, p = 0.86 — far above the 0.05 threshold, so normality cannot be rejected. The moment skewness coefficient is ≈ 0.14 (near zero), confirming the Cholesterol values are approximately symmetric with only a very slight right skew, unsurprising given the small, evenly-spread sample.

Question 3 (R)

Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk?

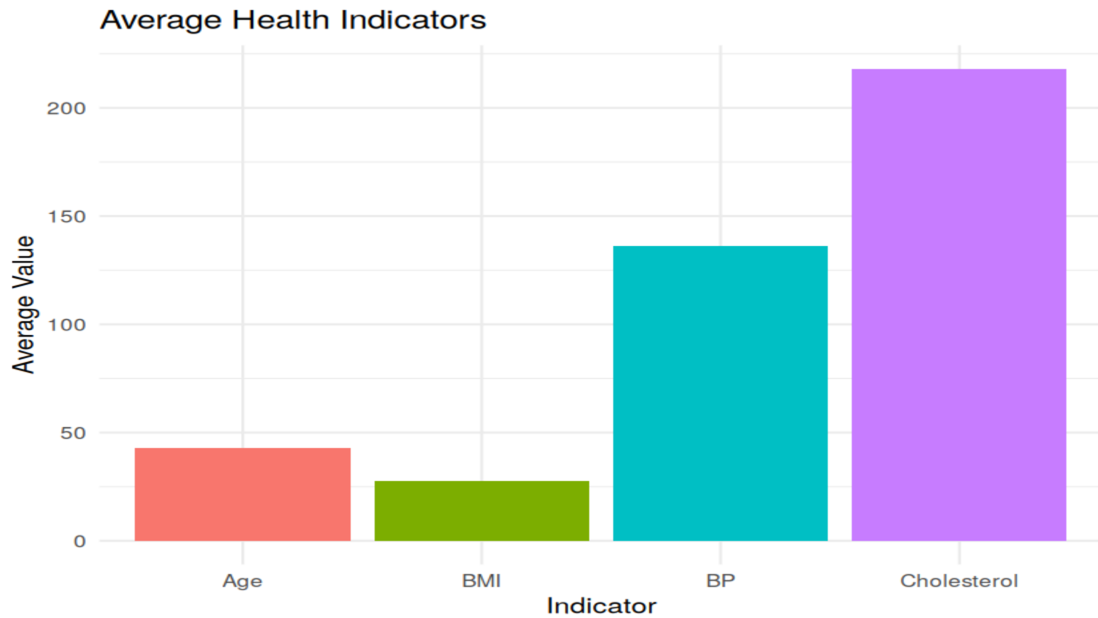
R Code:

```
library(tidyr); library(dplyr)

df27_avg <- pivot_longer(df27[,2:5], cols = everything(),
                         names_to = "Metric", values_to = "Value") %>%
  group_by(Metric) %>% summarize(Avg = mean(Value))

ggplot(df27_avg, aes(x = Metric, y = Avg, fill = Metric)) +
  geom_bar(stat = "identity", show.legend = FALSE) + theme_minimal() +
  labs(title = "Average Health Indicators", x = "Indicator", y = "Average Value")
```

Output — Question 3:



Interpretation:

Average Age is 43, BMI 27.6 (overweight range), BP 136 (borderline high), and Cholesterol 218 (borderline high). Taken together, the average patient in this cohort already sits in a borderline-risk zone, meaning even 'typical' patients here warrant monitoring, not just the oldest ones.

Question 4 (Tableau)

Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an Age filter for exploring risk patterns. Include hover tooltips showing all health indicators. Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.

Steps to perform in Tableau:

- Connect Tableau to Patient_Health.csv.
- Build a highlight table: Patient_ID on Rows, a dummy 'Risk' column on Columns, Cholesterol (or BP) on Color to create the heatmap.
- Drag Age to the Filters shelf as a range filter.
- Add Age, BMI, BP, and Cholesterol to Tooltip so hovering over a patient shows all four indicators.

Interpretation:

Analysis: P5 (age 60, BMI 32, BP 150, Chol 260) is clearly the highest-risk patient, and P3 (age 55) is second — both align with the strong age-risk correlation observed in Q1. Age is effectively a single proxy for overall risk in this dataset. A dashboard like this lets clinicians sort/filter by Age or Cholesterol at a glance and prioritize outreach to the P5/P3-type patients first.

Set 28: Vehicle Performance Analysis

Dataset: Vehicle Performance

Vehicle_ID	Engine_Size (L)	Horsepower	Fuel_Efficiency (km/l)	Top_Speed (km/h)	Safety_Rating
V1	1.5	110	18	180	4
V2	2.0	150	15	200	5
V3	3.0	250	12	250	5
V4	2.5	200	14	220	4
V5	1.8	130	17	190	3

Question 1 (R)

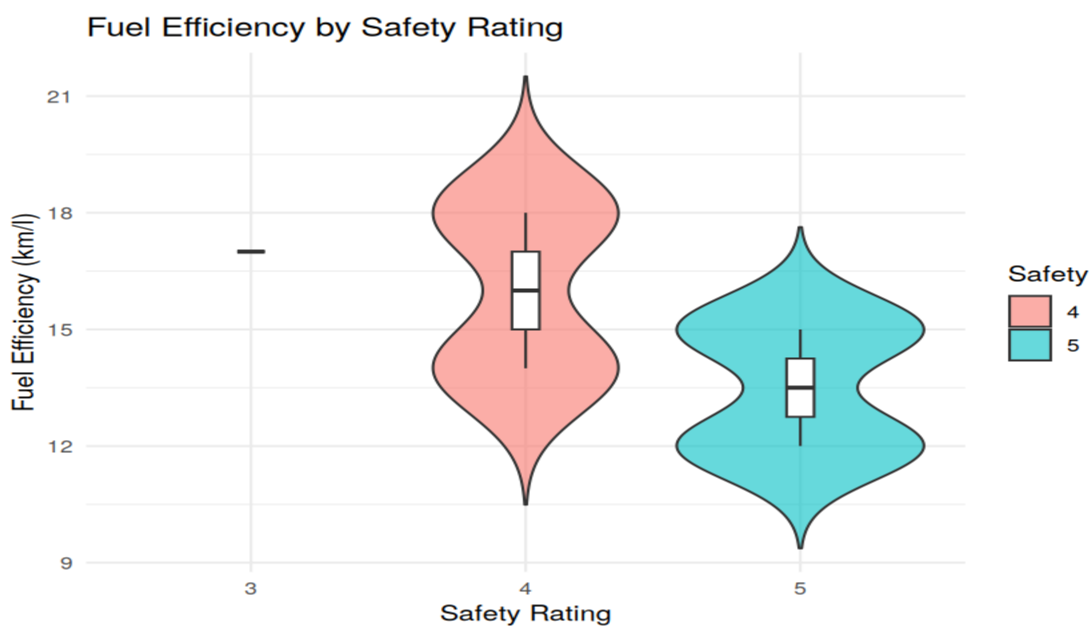
Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

R Code:

```
df28 <- data.frame(
  Vehicle_ID = c("V1","V2","V3","V4","V5"), Engine_Size = c(1.5,2.0,3.0,2.5,1.8),
  Horsepower = c(110,150,250,200,130), Fuel_Efficiency = c(18,15,12,14,17),
  Top_Speed = c(180,200,250,220,190), Safety_Rating = c(4,5,5,4,3)
)

ggplot(df28, aes(x = as.factor(Safety_Rating), y = Fuel_Efficiency, fill =
as.factor(Safety_Rating))) +
  geom_violin(trim = FALSE, alpha = 0.6) +
  geom_boxplot(width = 0.1, fill = "white") + theme_minimal() +
  labs(title = "Fuel Efficiency by Safety Rating", x = "Safety Rating",
       y = "Fuel Efficiency (km/l)", fill = "Safety")
```

Output — Question 1:



Interpretation:

Safety=4 vehicles show the widest spread (15-18 km/l), Safety=5 vehicles cluster lower (12-15 km/l), and Safety=3 has a single data point (17 km/l). The pattern hints that higher safety ratings tend to come with heavier/more powerful vehicles that are less fuel-efficient — a common efficiency-vs-safety trade-off.

Question 2 (R)

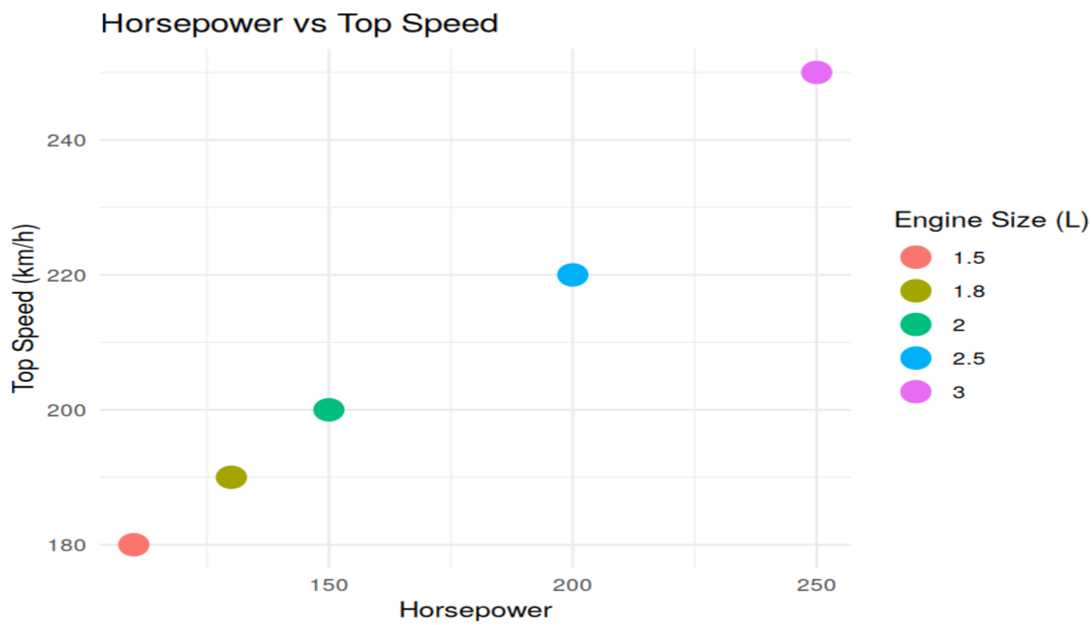
Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

R Code:

```
ggplot(df28, aes(x = Horsepower, y = Top_Speed, color = as.factor(Engine_Size))) +
  geom_point(size = 5) + theme_minimal() +
  labs(title = "Horsepower vs Top Speed", color = "Engine Size (L)",
       x = "Horsepower", y = "Top Speed (km/h)")
```

```
cor(df28$Horsepower, df28$Top_Speed)
```

Output — Question 2:



Interpretation:

Horsepower and Top Speed are almost perfectly correlated (r ≈ 0.997) and rise together with Engine Size — the 3.0L V3 has both the highest horsepower (250) and top speed (250 km/h). This is a strong, positive, near-linear relationship: bigger engines reliably deliver more horsepower and higher top speed.

Question 3 (R)

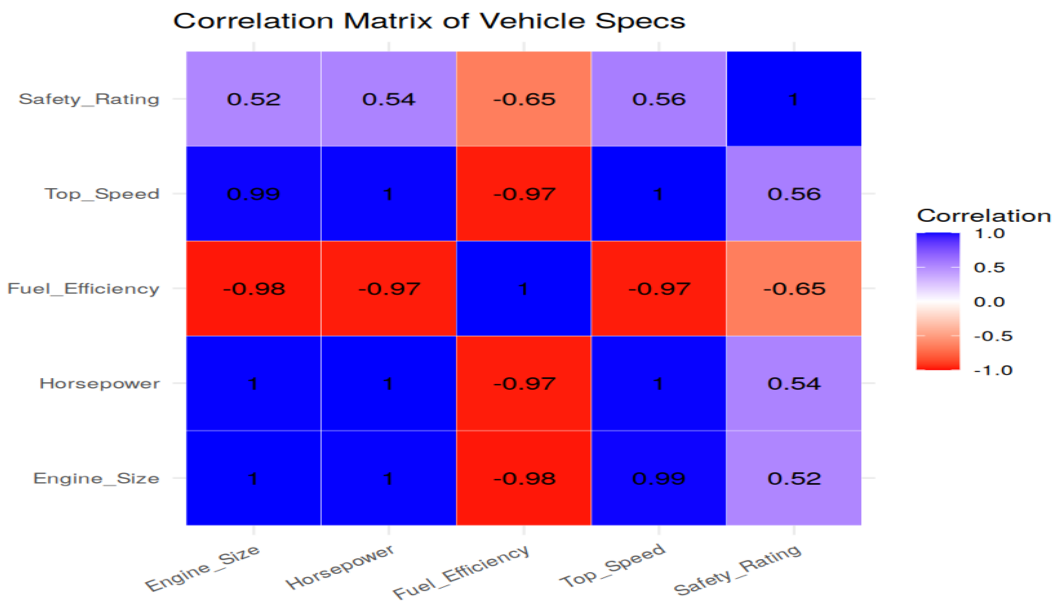
Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

R Code:

```
corr_matrix <- cor(df28[, 2:6])
corr_df <- as.data.frame(as.table(corr_matrix))
names(corr_df) <- c("Var1","Var2","Correlation")

ggplot(corr_df, aes(Var1, Var2, fill = Correlation)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(Correlation, 2)), size = 4) +
  scale_fill_gradient2(low = "red", mid = "white", high = "blue", midpoint = 0, limits = c(-
1,1)) +
  theme_minimal() + labs(title = "Correlation Matrix of Vehicle Specs", x = "", y = "")
```

Output — Question 3:



Interpretation:

Top_Speed is most strongly (positively) associated with Horsepower (r=1.00) and Engine_Size (r=0.99), and most strongly (negatively) associated with Fuel_Efficiency (r=-0.97). Safety_Rating has the weakest link to Top_Speed (r=0.56). In short: bigger engine -> more horsepower -> higher top speed -> lower fuel efficiency is the dominant performance chain in this dataset.

Question 4 (Tableau)

In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for Engine_Size and Safety_Rating, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.

Steps to perform in Tableau:

- Connect Tableau to Vehicle_Performance.csv.
- Sheet 1 (Bar): Vehicle_ID on Columns, Fuel_Efficiency on Rows.
- Sheet 2 (Scatter): Horsepower on Columns, Top_Speed on Rows, colored by Safety_Rating.
- Add Engine_Size and Safety_Rating to the Filters shelf on both sheets.
- Combine on a Dashboard.

Interpretation:

Interpretation: V5 (1.8L, 17 km/l, 190 km/h, Safety 3) stands out as the best efficiency-to-performance balance among the smaller-engine vehicles, while V3 (3.0L) sacrifices efficiency (12 km/l) for top-end performance (250 km/h). Filtering by Safety >= 4 removes V5, showing the real trade-off buyers face between safety, efficiency and speed in this fleet.

Set 29: Student Academic Performance

Dataset: Student Academic Performance

Student_ID	Age	Study_Hours	Attendance (%)	Test_Score	Participation_Score
S1	19	12	90	85	8
S2	21	8	70	70	7
S3	20	15	95	92	9
S4	22	10	85	80	8
S5	23	7	60	65	6

Question 1 (R)

Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

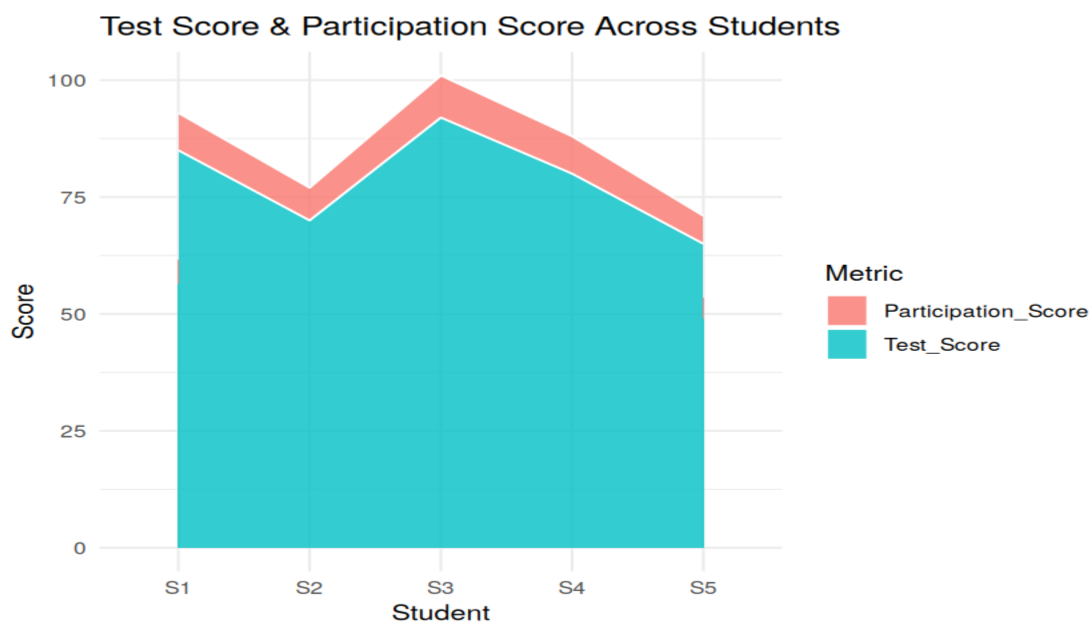
R Code:

```
df29 <- data.frame(
  Student_ID = c("S1","S2","S3","S4","S5"),
  Age = c(19,21,20,22,23),
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60),
  Test_Score = c(85,70,92,80,65),
  Participation_Score = c(8,7,9,8,6)
)

df29_long <- pivot_longer(df29, cols = c(Test_Score, Participation_Score),
                          names_to = "Metric", values_to = "Value")
df29_long$Student_ID <- factor(df29_long$Student_ID, levels = df29$Student_ID)

ggplot(df29_long, aes(x = Student_ID, y = Value, fill = Metric, group = Metric)) +
  geom_area(alpha = 0.8, color = "white") + theme_minimal() +
  labs(title = "Test Score & Participation Score Across Students", x = "Student", y = "Score")
```

Output — Question 1:



Interpretation:

Both Test_Score and Participation_Score rise and fall together across students (S3 highest on both, S5 lowest on both), reinforcing that students who participate more in class also tend to score higher on tests.

Question 2 (R)

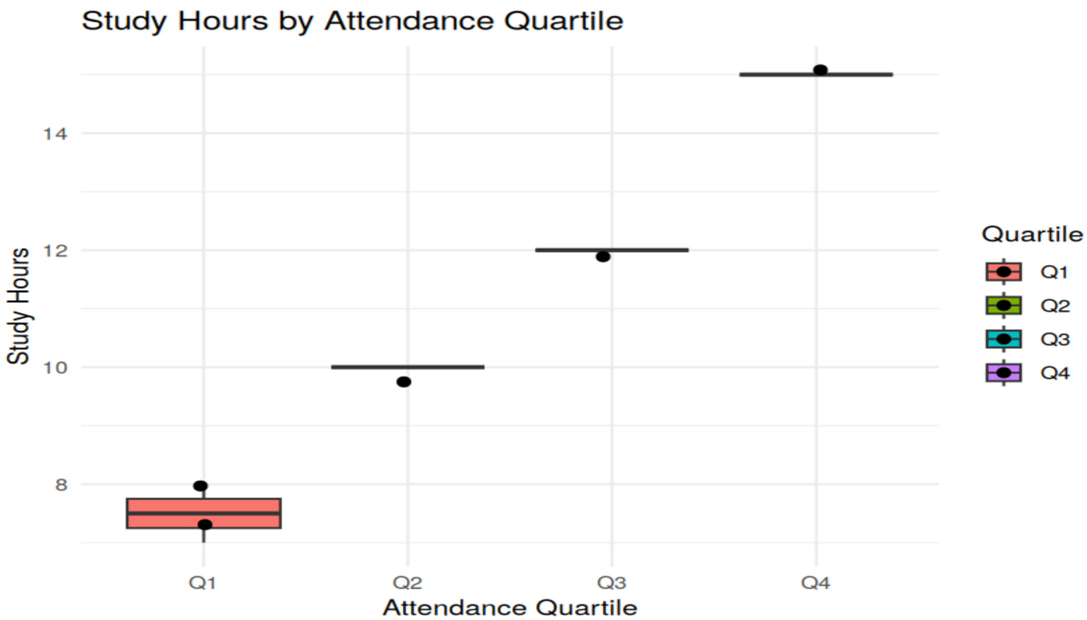
Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

R Code:

```
brks <- quantile(df29$Attendance, probs = seq(0,1,0.25))
df29$Attendance_Quartile <- cut(df29$Attendance, breaks = unique(brks),
                                include.lowest = TRUE,
                                labels = paste0("Q", 1:(length(unique(brks))-1)))
```

```
ggplot(df29, aes(x = Attendance_Quartile, y = Study_Hours, fill = Attendance_Quartile)) +
  geom_boxplot() + geom_jitter(width = 0.05, size = 2) + theme_minimal() +
  labs(title = "Study Hours by Attendance Quartile", x = "Attendance Quartile",
       y = "Study Hours", fill = "Quartile")
```

Output — Question 2:



Interpretation:

With only 5 students, each attendance quartile contains just one or two students, so each box collapses to a single value rather than a true spread: Q1 (S2, S5: 7-8 hrs), Q2 (S4: 10 hrs), Q3 (S1: 12 hrs), Q4 (S3: 15 hrs). No outliers are flagged, and study hours increase steadily from Q1 to Q4 — students with higher attendance also tend to log more study hours.

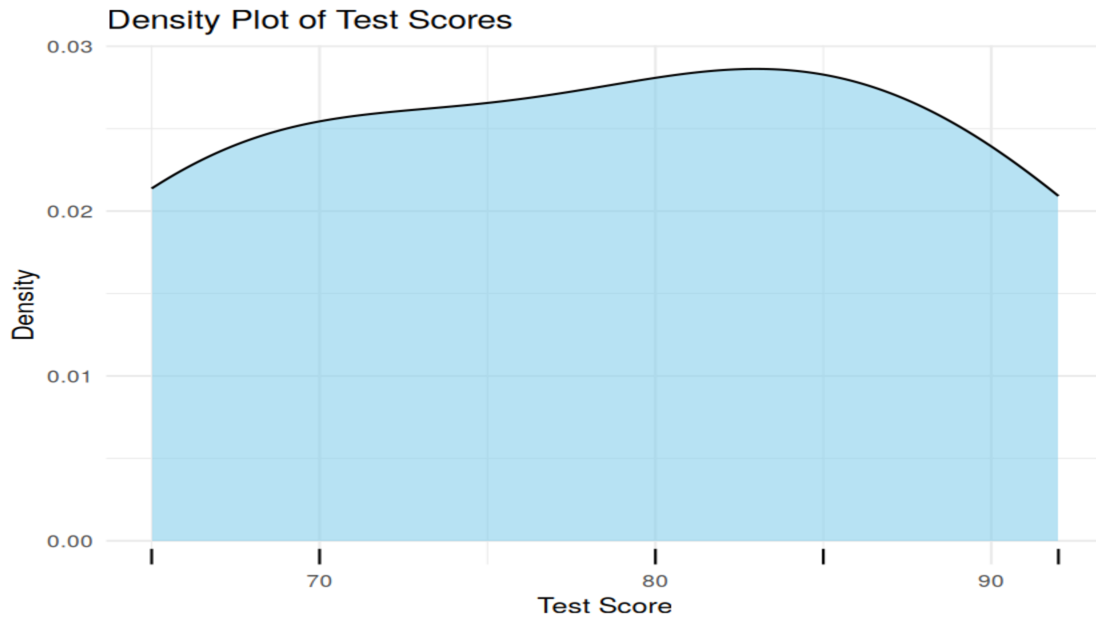
Question 3 (R)

Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

R Code:

```
ggplot(df29, aes(x = Test_Score)) +
  geom_density(fill = "skyblue", alpha = 0.6) + geom_rug() + theme_minimal() +
  labs(title = "Density Plot of Test Scores", x = "Test Score", y = "Density")
```

Output — Question 3:



Interpretation:

Test scores range 65-92 (mean ≈ 78.4) with a single broad peak around the low-80s and no separate cluster or long tail — no outliers are apparent in this small, fairly evenly-spread sample.

Question 4 (Tableau)

In Tableau, design an interactive scatterplot with Study_Hours on X-axis, Test_Score on Y-axis, colored by Participation_Score. Include filters for Age.

Steps to perform in Tableau:

- Connect Tableau to Student_Academic_Performance.csv.
- Drag Study_Hours to Columns and Test_Score to Rows; uncheck 'Aggregate Measures'.
- Drag Participation_Score to Color.
- Drag Age to the Filters shelf as a range filter and set it to interactive (show filter card).

Set 30: Mobile App Usage (Mobile_App_Usage.csv)

Dataset: Mobile_App_Usage.csv

User_ID	Gender	Age	Screen_Time (hrs)	App_Usage_Count	Data_Used (GB)	Satisfaction	Usage_Date
U01	Male	20	4.5	18	2.4	3	2025-01-08
U02	Female	22	6.0	25	3.8	5	2025-01-08
U03	Male	19	3.2	12	1.6	3	2025-02-11
U04	Female	21	7.1	30	4.5	5	2025-02-11
U05	Male	23	2.8	10	1.2	2	2025-03-14
U06	Female	20	5.4	22	3.1	4	2025-03-14

Question 1 (R)

Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

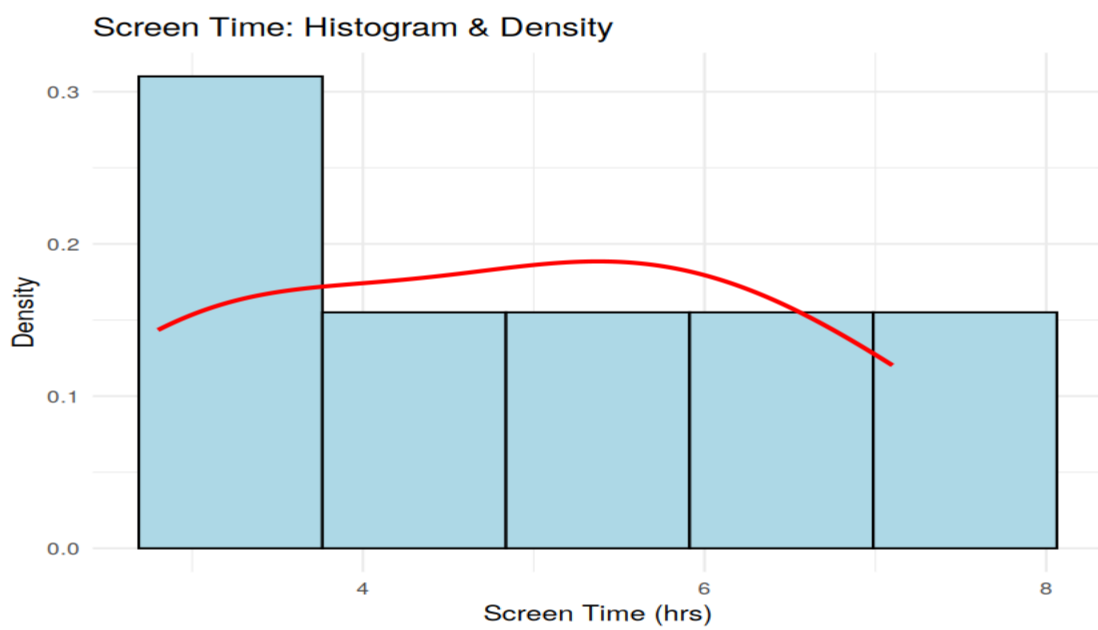
R Code:

```
df30 <- data.frame(
  User_ID = c("U01","U02","U03","U04","U05","U06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
```

```
    Age = c(20,22,19,21,23,20),
    Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
    App_Usage_Count = c(18,25,12,30,10,22),
    Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),
    Satisfaction = c(3,5,3,5,2,4),
    Usage_Date = as.Date(c("2025-01-08","2025-01-08","2025-02-11","2025-02-11","2025-03-
  14","2025-03-14"))
  )

  ggplot(df30, aes(x = Screen_Time)) +
    geom_histogram(aes(y = after_stat(density)), bins = 5, fill = "lightblue", color = "black") +
    geom_density(color = "red", linewidth = 1) + theme_minimal() +
    labs(title = "Screen Time: Histogram & Density", x = "Screen Time (hrs)", y = "Density")
```

Output — Question 1:



Interpretation:

Screen time ranges from 2.8 to 7.1 hours with no strong single peak in this small sample — usage is fairly spread out, with a slight concentration around 4.5-5.4 hours.

Question 2 (R)

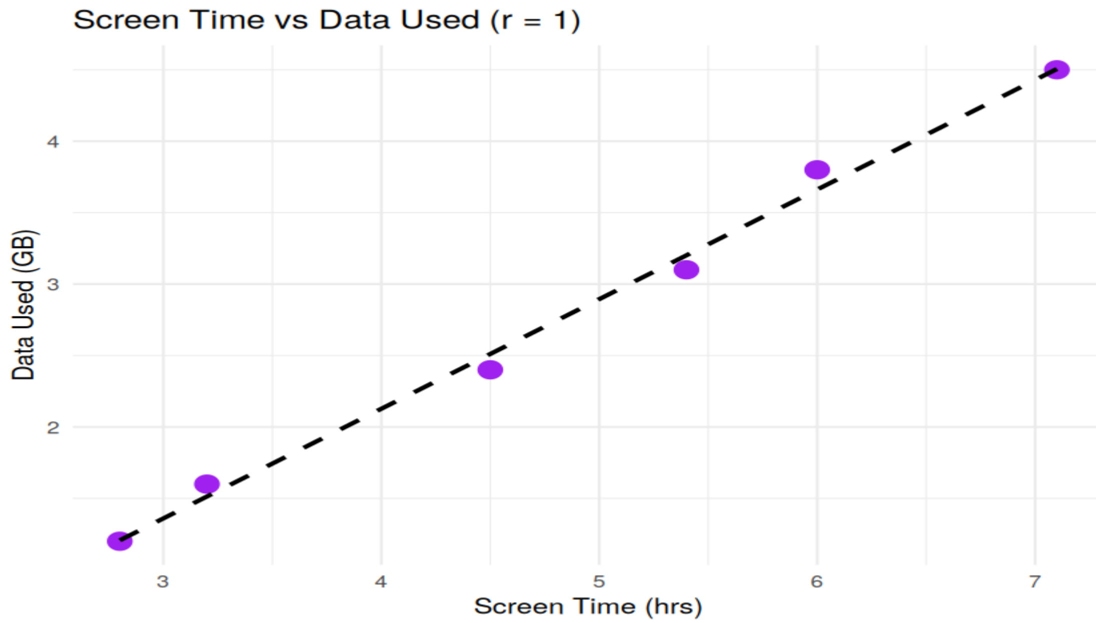
Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

R Code:

```
  cor(df30$Screen_Time, df30$Data_Used)

  ggplot(df30, aes(x = Screen_Time, y = Data_Used)) +
    geom_point(size = 4, color = "purple") +
    geom_smooth(method = "lm", se = FALSE, color = "black", linetype = "dashed") +
    theme_minimal() + labs(title = paste0("Screen Time vs Data Used (r = ",
  round(cor(df30$Screen_Time, df30$Data_Used),2), ")"),
                            x = "Screen Time (hrs)", y = "Data Used (GB)")
```

Output — Question 2:



Interpretation:

Correlation r ≈ 0.997 — an almost perfectly linear relationship. Every additional hour of screen time tracks closely with a proportional rise in data consumption, which makes intuitive sense (more time on apps = more data pulled).

Question 3 (R)

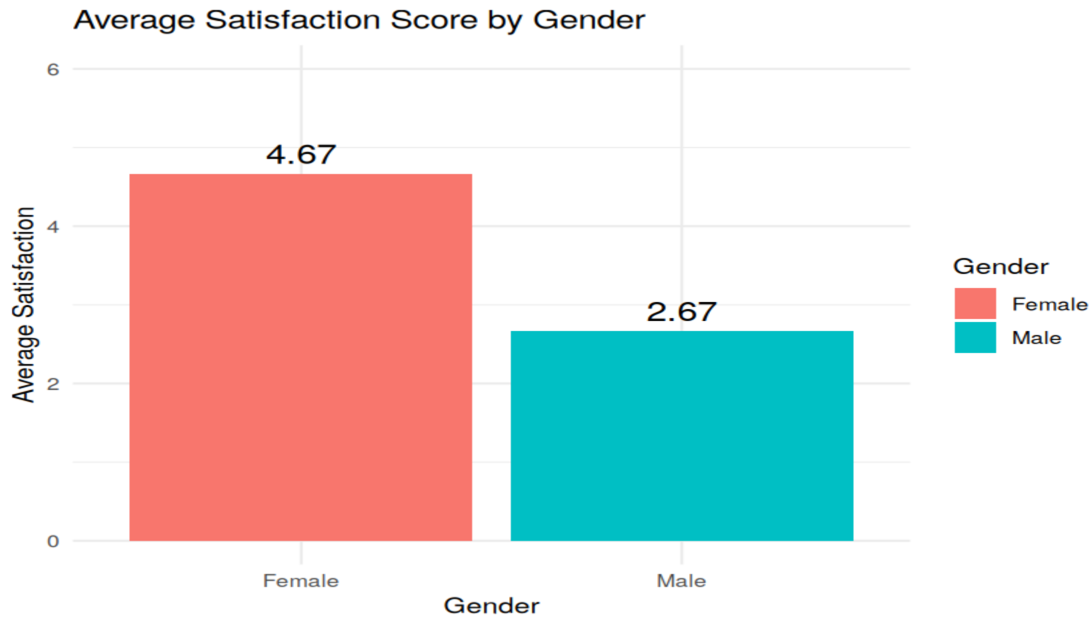
Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

R Code:

```
df30_avg <- df30 %>% group_by(Gender) %>% summarize(Avg_Satisfaction = mean(Satisfaction))

ggplot(df30_avg, aes(x = Gender, y = Avg_Satisfaction, fill = Gender)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = round(Avg_Satisfaction,2)), vjust = -0.5, size = 5) +
  theme_minimal() + ylim(0, 6) +
  labs(title = "Average Satisfaction Score by Gender", x = "Gender", y = "Average
Satisfaction")
```

Output — Question 3:



Interpretation:

Average satisfaction is notably higher for Female users (4.67) than Male users (2.67) in this sample. This lines up with Female users also logging higher average Screen_Time and Data_Used, suggesting engagement and satisfaction move together here.

Question 4 (Tableau)

Import Mobile_App_Usage.csv into Tableau. Create a bubble chart (Screen Time vs Data Used). Use color for Gender and size for App Usage Count. Create a line chart for usage trends. Add filters for Gender and Usage_Date. Design a mini dashboard.

Steps to perform in Tableau:

- Connect Tableau to Mobile_App_Usage.csv.
- Sheet 1 (Bubble): Screen_Time on Columns, Data_Used on Rows, change mark to Circle; Gender on Color, App_Usage_Count on Size.
- Sheet 2 (Line): Usage_Date on Columns (continuous), SUM(App_Usage_Count) or AVG(Screen_Time) on Rows.
- Add Gender and Usage_Date to the Filters shelf on both sheets.
- Combine both sheets on a Dashboard.